

Language and brain-to-brain (B2B) communication

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ABSTRACT. We review the first proof-of-principle demonstration for successful non-invasive brain-to-brain communication of conscious thought without the use of input-output systems of articulation and auditory perception of speech. In its present stage, the proof is overstated. First, the communication transmitted is not structured thought but a consciously perceived digital code. Second, communication is transmitted with the use of extralinguistic input-output mechanisms of motor and occipital cortices. It may be an innovative and potentially useful technological development but it is without explanatory significance for the study of language in the brain.

Key words: brain-to-brain communication, non-invasive technology, digital code, internal language, motor activation, phosphene perception, Chinese room

[1] The capacity for language uniquely provides humans with the computational means for internally generating an unbounded array of hierarchically structured expressions with determinate compositional meanings (Chomsky, 2016; 2020; 2021). The capacity must be sharply distinguished from the use of language in speech or sign. However, failure to distinguish between an internal language mechanism, a biological system coded in the brain, and the communicative use of that system in externalized speech or sign is widespread in studies of language and evolution of language. Even apparent exceptions to this generalization sometimes break down on closer inspection. Studies suggesting the feasibility of non-invasive brain-to-brain (B2B) communication of conscious thought without the use of language come to mind (Grau et al., 2014; Rao et al., 2014; Pereira et al., 2018; Jiang et al., 2019). In these studies, which advance Miguel Nicolelis’s ground-breaking research on the brain-computer interface with monkeys (Nicolelis, 2010), conscious thought is conceived of as being distinct from language, which only exists in the external use of it.

[2] A recent technological experiment from cognitive neuroscience claims to provide a “critical proof-of-principle demonstration for the development of conscious brain-to-brain communication” (Grau et al., 2014). Using high-tech systems, conscious thought is said to be transmitted from one brain to another without the use of (externalized) language, that is, without communication through standard sensorimotor modalities. Encrypted messages encoding, e.g., the Catalan word *hola* — or, in principle, any phrase or sentence of any language, e.g., Chinese *wǒ shǒu xiě wǒ kǒu* — can be transmitted from brain to brain. Transmission is executed by combining a brain-computer interface (BCI) with a computer-brain interface (CBI) and applying transcranial magnetic stimulation (TMS) technologies without intervention of the peripheral sensorimotor systems for vocal articulation and auditory perception. Language is viewed here as the ancillary use of a cognitive system, not the internal system that generates structured thought. In this sense, B2B communication of internal language may be said to bypass the use of externalized language, speech or sign.

[3] However, upon closer examination it turns out that what is emitted is not conscious thought but voluntary motor imagery-controlled EEG data and what is received is something else, conscious phosphene perceptions. For electronic transmittal of information, the emitter encodes the bit values “1” and “0” of the binary code through motor imagery of hands and feet respectively. The phosphene sensations of the receiver represent “active” and “silent” directions of TMS-induced pulses that code the bit values “1” and “0” respectively. Movement of hands/feet thus correlate with active/silent phosphene sensations. The conscious mind is only

aware of a binary code to act upon, not of the encrypted underlying thought. The experiment directly links motor imagery in the motor cortex of one brain to phosphene perceptions in the occipital cortex of another brain. The minds are ignorant of the meaning coded in the EEG/coil information that is processed by motor/occipital cortices. The latter only serve as input-output channels for articulation and perception of binary digits that have no access to (and cannot be accessed by) central computational systems that generate compositional meaning of internal language. The cortical systems communicate but the communicators have no idea what it is that they communicate about. Impossible language constructs (“a tree seems he not”) could equally well receive a binary code that is successfully B2B communicated. From a cognitive neuroscience point of view, failure or indifference to distinguish between possible and impossible language constructs is already indication of the experiment’s futility. There’s no B2B communication of propositional content, i.e., internal language or thought.

[4] Neither motor cortex (of emitter) nor occipital cortex (of receiver) connects to these internal language systems, the “conscious” part of communication being restricted to conscious *non-linguistic* motor and sensory activity. Consequently, the result is non-standard communication of non-linguistic information rather than transmission of meaningful linguistic information without external signaling. Communication is neither communication of internal language nor communication without external use of peripheral systems. Motor activation and phosphene perception simply substitute for the vocal articulation and auditory perception systems of language use. And the message transmitted is a visually perceived digital code, not internal language or propositional content. Rather the binary code, which is not language-specific, only receives linguistic meaning after the code is explained to each mind. By nature, I-language generates the “code” (the converging merge-based derivational construct) for the language-internal interpretive and language-external sensorimotor systems to process and act upon. In the experiment, I-language is disconnected from the input-output channels. Peripheral brain systems communicate binary digital information but the mind is ignorant of its semantic contents. The internal language system would be of no help. There’s simply exportation of a binary code encoding motor imagery and importation of a binary code for TMS-pulses.

[5] On one interpretation the demonstration is reminiscent of Searle’s famous *Chinese Room* argument (Searle, 1980) that no digital computer running a program can have a mind consciously understanding the program. See Figure 1. Leaving aside the question of whether machines can think (Turing 1950, Searle 1980) or have consciousness (Searle, 1997; Chomsky, 2018; Penrose, 1989), the argument adapted to brain-to-brain communication with BCI and CBI technologies shows that communication is essentially divorced from language. Motor cortex activations, which are not caused by internal language, are transmitted, and perceived elsewhere as phosphene sensations in the occipital cortex, which are not fed either into the internal language system. Neither emitter nor receiver can relate motor activity or sensory experience to linguistic meaning. Analogous to Searle’s Chinese Room, there is “blind” interaction between ancillary/peripheral input-output systems that have no access to central systems of internal language. The experiment forgets about the mind that should hold the key to the code (the internal language), restricting itself to the brain areas that produced/exported the externalized information (motor cortex activation) and the brain areas that processed/imported the external signal (occipital cortex stimulation). Communication is between cortical sensorimotor mechanisms but without understanding from the internal language system. In effect, communication of meaningless symbols.

[6] In the Chinese Room thought experiment the man inside the room, who is running the program without understanding its meaning, may be thought of as the computational “brain,” while the whole system itself may be interpreted as a model of the “mind” (the Berkeley interpretation). In the new setting, the program is run by BCI/ICB technologies that connect motor activation of one brain to occipital phosphene perceptions of another. The motor activation and phosphene perceptions encode a message that even the mind doesn’t understand unless it is given access to the code for computing the information. The brain in the experiment is dissociated from the mind that doesn’t cognize the program run by the brain.

Extended Chinese Room Argument

after Searle's "Minds, brains, and programs" (Searle, 1980)

At one location a person P1 (who doesn't speak Chinese) receives a binary code encoding a "Chinese message" which he wouldn't understand – *wǒ shǒu xiě wǒ kǒu* (我手寫我口). Internet-mediated B2B communication with another person P2 (who also doesn't speak Chinese) at a second location is established by combining a Brain-Computer Interface (BCI) with a Computer-Brain Interface (CBI). BCI is based on voluntary motor imagery-controlled *electroencephalographic* information; CBI induces the conscious perception of *phosphenes* (light flashes) through neuro-navigated, robotized *transcranial magnetic stimulation*. The two brains process digital information, executing a mental program they don't "cognize." There is blind brain-to-brain interaction between input-output systems without any access to an internal language system for constructing/interpreting "audible signs for thought" (William D. Whithney).

In this case, access to such a central system would even be useless. P1 and P2 only process symbols and perceptions that encode an expression – *wǒ shǒu xiě wǒ kǒu* – which is meaningful to other minds, i.e., native speakers of Chinese, but not to P1 or P2. To interpret the message, the conscious phosphene perceptions must be decoded again into the original Chinese message and must, for further understanding by P1 and P2, be translated into whatever language is internally represented in their brains.

For speakers of Chinese, the predicate *xiě* ("write") takes the NP *wǒ kǒu* ("my mouth") as its theme argument to yield the VP *xiě wǒ kǒu*, which combines with the NP *wǒ shǒu* ("my hand") as its agent argument, thus yielding the proposition S – [s [NP *wǒ shǒu*] [VP [v *xiě*] [NP *wǒ kǒu*]]] – with its compositional meaning "my hand writes [what] my mouth [says]." In this experiment, communication is between cortical sensorimotor mechanisms but without propositional understanding from an internal language system. The experiment forgets about the mind that holds the key to the code transmitted.

Note: The expression *wǒ shǒu xiě wǒ kǒu* was taken from Rint Sybesma, interviewed by Berthold van Maris, "Het rijk van de duizend talen," *NRC Wetenschap*, 13 May, 2017, W10-W11.

Figure 1. Extended Chinese Room argument, after Searle's "Minds, brains, and programs" (Searle, 1980)

[7] Transmission of (structured) thought without language as a mediating factor is an illusion. First, structured thought is, or is co-extensive with, hierarchically structured language, I-language, generated by an internal recursive procedure. Therefore, any B2B/M2M communication of thought must be communication of internal language. B2B/M2M communication skipping internal language cannot be a viable goal to pursue. Next, ancillary sensorimotor systems are only used for externalization (vocal articulation) of internal language or processing (parsing) of externalized language. It's true that the experiment skips the peripheral input-output systems for externalizing language and processing speech, but only because an analogous pair of input-output channels, namely motor and occipital cortices, are substituted for the latter. Finally, what's peripherally communicated is basically a binary code without propositional content. The mind has no conscious knowledge of its meaning. The coded thought does not reach the internal language systems and cannot be accessed by them. It only has a meaning to an outsider mind holding the key. There's no meaningful communication.

[8] The B2B/M2M communication reported in the study reviewed here is not communication of propositional content. Communication only handles input-output exchange of encrypted messages in a binary code whose meanings are not accessible to internal systems of language and thought. Emitters/receivers don't know the code nor what it represents or how the encoded/decoded message enters into interpretation. Regardless of all this, a more principled approach to B2B that incorporates a recent result of the study of language in the brain may bring a solution a step closer. There is intriguing evidence that acoustic information of phonological

structure must be internally represented as I-language in non-acoustic areas, such as Broca's area (Magrassi et al., 2015). Interestingly, the electrocorticogram (ECoG) correlates with the sound envelope of the utterance, *even in the absence of speech*. Plausibly, the message decoded from ECoG would be an amalgam of I-language and the peripheral sensory motor systems that yield externalized structure (Chomsky 2016, 2020, 2021), perhaps something like $[_{VP}[_{NP} w\check{o} sh\check{o}u] [_{VP} xi\check{e} [_{NP} w\check{o} k\check{o}u]]]$. Possibly, using BCI/CBI technology, extracted ECoG information could be digitally encoded, brain-to-brain transmitted, decoded, and fed into the cortical language centers to be accessed for semantic interpretation by internal language. In the present case, this would be the I-language system of the Chinese mind/brain, which ultimately yields the compositional meaning “my hand writes [what] my mouth [says],” the poet-diplomat Huáng Zūnxiàn's slogan to promote writing in the vernacular. Again, there's no way of circumventing internal language in B2B or M2M communication. After all, I-language is essential for structured thought. But integrating I-language requires substantial theoretical advances towards “the ultimate goal of neuroscience,” which is “to mechanistically link the physical elements of the brain to the more ethereal phenomena of the mind” (Bizzi and Ajemian, 2015). A hard problem to solve.

[9] The ability to converse directly, “brain-to-brain” or “mind-to-mind,” skipping language altogether has been argued by proof-of-concept demonstration to be a viable prospect but without much theoretical understanding. Cortical activation in the language areas, to state the obvious, cannot be causally interpreted without some proper understanding of the neural code used for generating language. Even if we were to know the “true” internal grammar, we would still need to know how that grammar is distributed across neural circuits. Because of this, direct B2B/M2M communication of freely generated structured thought without the need to access ancillary SM modalities must be a hope for the future rather than a present reality. In agreement with MIT neuroscientist Robert Ajemian [p.c.] we may conclude that the attempt to develop conscious brain-to-brain communication seems “a-priori methodologically flawed and not simply flawed because of any specific scientific assumptions.” Perhaps, for the distant future, the best approximation to B2B may be exchanges of electrocortical information between the cross-subject frontal and temporal language areas that are implemented in externalization and processing of linguistic input, each with direct access to the primary areas of internal language. If electrocorticograms contain sufficient informational structure (e.g., instructive amalgams of I-language and SM properties), then a parsing function $f^1(ECoG, MRC)$ may inversely apply Merge to yield conceptual-intentional structure with propositional content. In his article “The science of mind reading,” James Somers speculates that two persons wearing a “thinking hat” equipped with BCI/CBI technology might “skip [external] language altogether, conversing directly, mind to mind.” (*The New Yorker*, Nov. 29, 2021). Possibly, good news for “locked-in” patients. But that's all in the future. Perhaps.

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